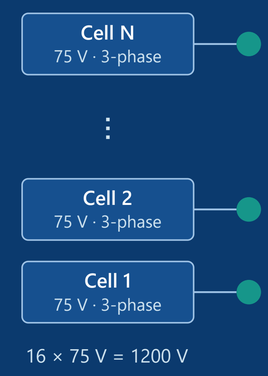
Moore4Power General Assembly · November 2026

Stacked Polyphase Bridge Traction Drives: Iterative GaN Building Block Design

Enes Ayaz, Shahriar Sarmast Ghahfarokhi, Kevin Klarin, Staffan Norrga

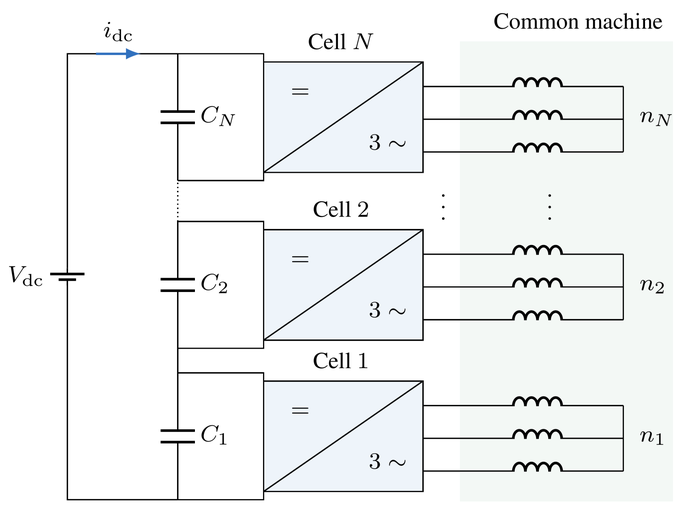
Department of Electric Power and Energy Systems, KTH Royal Institute of Technology, Stockholm, Sweden



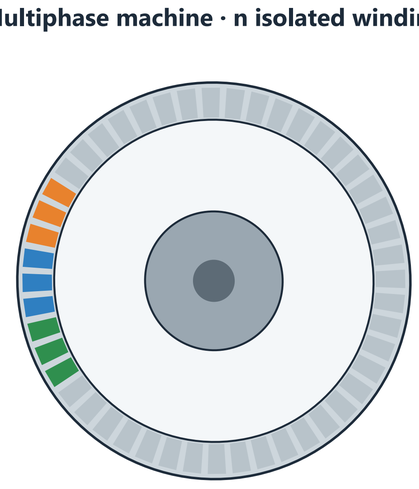
1200 V	16 × 75 V	4 × GaN devices	0.41 nH	99.71 %	2.85 %
dc link, 300 kW traction	series cells, own winding groups	100 V GaN per switch	commutation loop per cell	semiconductor efficiency	shared-bus dc ripple, 50 kHz

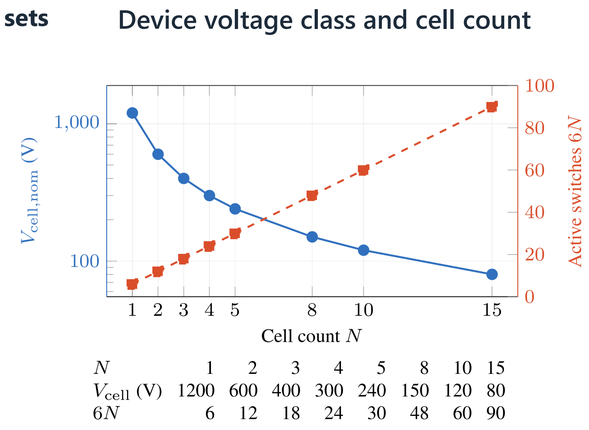
A · Review: what the SPB architecture demands

1 Concept: split the dc voltage, keep one machine



N-cell SPB: series dc links, separate winding groups



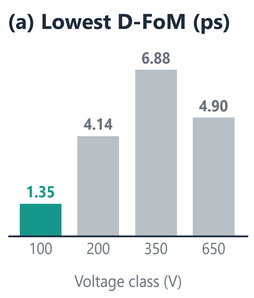


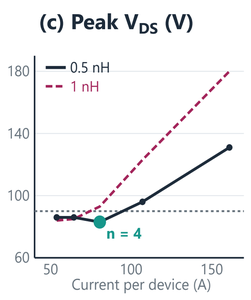
N	1	2	3	4	5	8	10	15
V_{cell} (V)	1200	600	400	300	240	150	120	80
$6N$	6	12	18	24	30	48	60	90

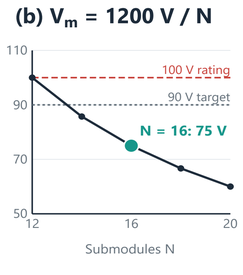
1200 V link: $V_{cell} = V_{dc}/N$ versus active switches $6N$

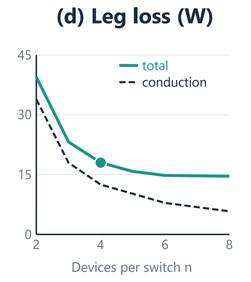
- N two-level bridges in series on the dc side; each feeds its own three-phase winding group.
- $V_{dc,k} \approx V_{dc}/N$: the device voltage class follows the cell, not the full link.
- All cells carry the same dc current; $P_{ac} = \Sigma P_k$, so cell voltages must be balanced actively.
- 6N switches, N drivers and capacitors: complexity moves to the system, it does not vanish.
- Rating with margin: $V_{rated} \geq k_{rel}(V_{dc,max}/N + \Delta V_{imb} + \Delta V_{sw})$.
- N = 16 gives 75 V cells for 100 V GaN; the overshoot L·di/dt sets the usable margin.

2 Device voltage class-cell count and single-cell screen





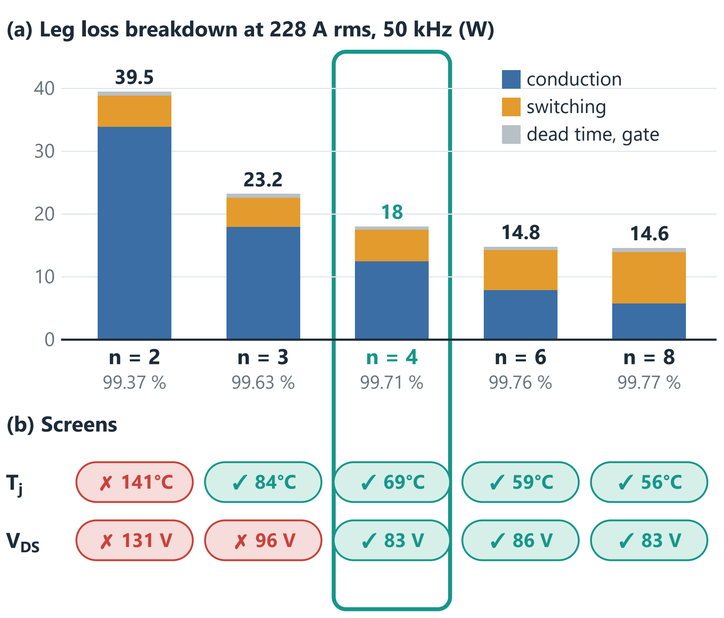




Lower blocking voltage raises the device figure of merit (D-FoM)

Four devices: 18 W per leg, the knee of the loss curve

3 Why four devices per switch

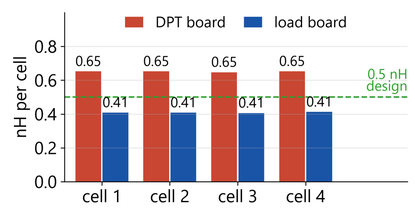


(b) Screens

	n = 2	n = 3	n = 4	n = 6	n = 8
T_j	✗ 141°C	✓ 84°C	✓ 69°C	✓ 59°C	✓ 56°C
V_{DS}	✗ 131 V	✗ 96 V	✓ 83 V	✓ 86 V	✓ 83 V

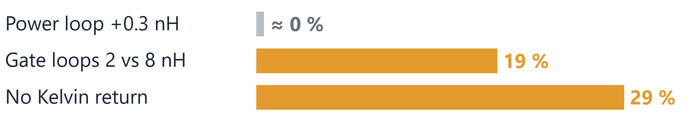
Limits: $T_j \leq 125$ °C, $V_{DS} \leq 90$ V at 0.5 nH. n = 4 is the smallest passing count.

4 Extracted parasitics (Ansys Q3D)



- Loop per cell: **0.41 nH** load board, 0.65 nH DPT (shunt); cells within 1.6 %
- Cell coupling k ≤ 0.034
- Gate loop, pair driven: 20.6 nH HS, 12.3 nH LS

(a) Switching-energy imbalance, two devices

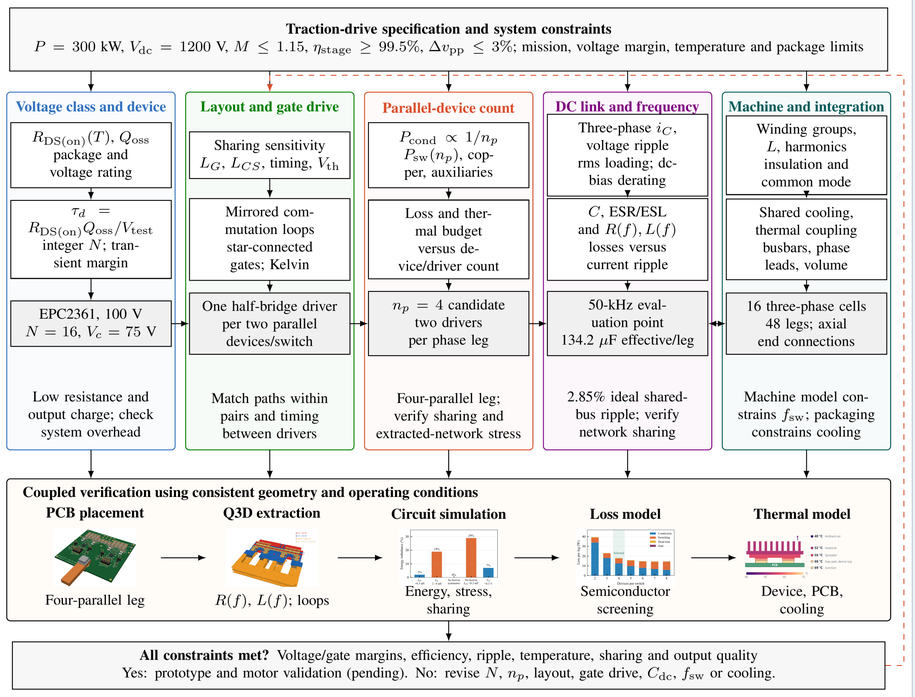


(b) Gate-loop inductance, Q3D (nH)

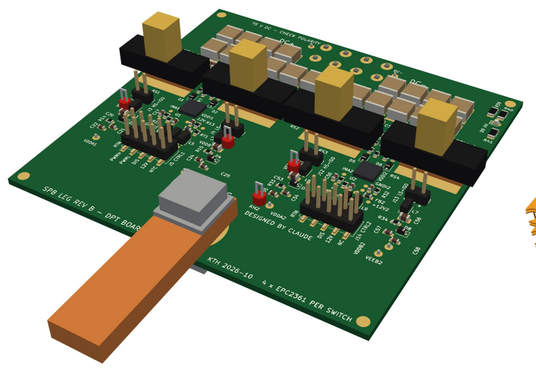


→ One 2EDF7275K channel per device pair, Kelvin returns

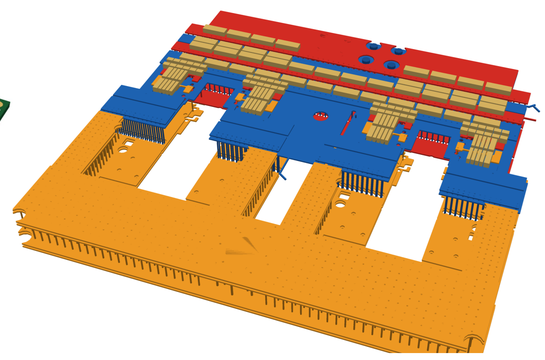
5 Iterative system design



6 Building block: double-pulse-test board and extracted network



DPT board: four cells, eight EPC2361



Ansys Q3D model of the four cells

0.41 / 0.65 nH
loop per cell, load / DPT

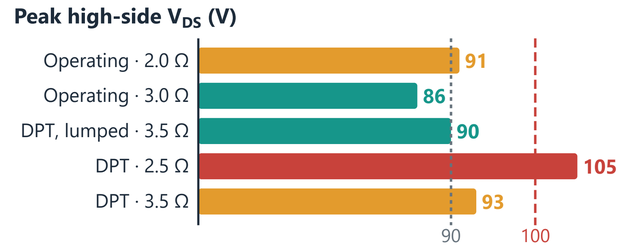
20.6 / 12.3 nH
gate loop, HS/LS pair

DC+
DC−
switch node
FET / MLCC

LTspice with the extracted network (75 V, 322 A)

- High-side v_{DS} peak **93 V** < 100 V rating
- Four parallel high-side gate loops equal within 0.3 %

Peak high-side V_{DS} (V)



✓ Conclusions

1. **16 × 75 V submodules let 100 V GaN** (EPC2361 lowest D-FoM) serve a 1200 V, 300 kW drive.
2. **Four devices per switch: 18 W per leg (99.71 %),** 69 °C junction, 83 V peak at 0.5 nH power loop inductance.
3. **The extracted gate loop (20.6 nH) sets the margin:** 3.5 Ω gate resistance during turn-on and −1.9 V off-state gate-source voltage give 93 V drain-source.

8 Project status and next steps

- DPT board, waiting PCB manufacturing and assembly
- Double-pulse tests at 75 V up to 322 A: per-cell current sharing, overshoot, E_{on}/E_{off}, gate bump.
- Three load-test legs + PCB busbar: one 75 V three-phase level, continuous and thermal tests.

9 Open research questions

- Optimum N: device class versus component count shifts with SiC, GaN and cooling.
- Joint battery–converter–machine design on vehicle energy, mass, cost and reliability.
- Stack-position CMV, bearing currents and insulation across the full stack.